

Improving Production Planning through Material Flow Analysis: Gas Stove Parts Manufacturer

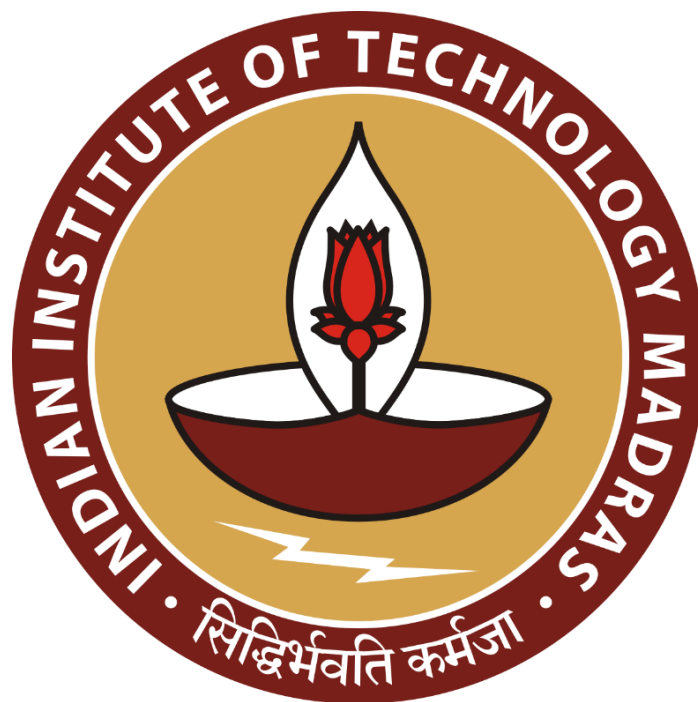
A Proposal report for the BDM capstone Project

Submitted By-

Name- Rishit Jain

Roll Number- 24F2004595

Email- 24F2004595@ds.study.iitm.ac.in



**IITM Online BS Degree Program,
Indian Institute of Technology,
Madras, Chennai Tamil Nadu, India,
600036**

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Declaration Statement

I am working on a Project Title “**A Data-Driven Framework for Improving Production Planning through Material Flow Analysis: Gas Stove Parts Manufacturer**”. I extend my appreciation to **Metafab India**, for providing the necessary resources that enabled me to conduct my project.

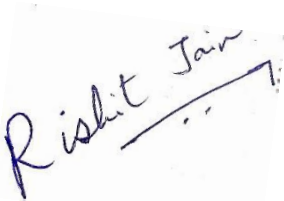
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.



Signature of Candidate: (**Digital Signature**)

Name: Rishit Jain

Date: 30th June 2026

1. Executive Summary and Title (200 Words)

This project focuses on **Metafab India**, a B2B manufacturing enterprise located in Greater Noida, Uttar Pradesh, engaged in the production of precision gas stove components including brass burner rings, cast iron pan supports and gas valve assemblies.

Preliminary examination of six months of operational records indicates the need to evaluate whether the organization's existing production planning practices effectively synchronize raw material procurement, manufacturing output and customer demand. Since production planning decisions directly influence inventory utilization, working capital requirements and order fulfilment, inefficient coordination among these operational activities may adversely affect overall business performance.

This study proposes a data-driven investigation using six months of primary business data comprising procurement records, production records and sales transactions. Additional operational records such as inventory, machine downtime and labour attendance will only be utilized where necessary to explain significant operational variations observed during analysis.

The project will employ descriptive analytics, material flow analysis, trend analysis and operational performance evaluation to identify opportunities for improving production planning decisions. The expected outcome is a practical production planning framework that enables MetaFab India to improve procurement decisions, optimize resource utilization and support evidence-based managerial decision-making.

2. Organization Background (150 Words)

MetaFab India is a privately owned B2B manufacturing enterprise operating from the Site-V Industrial Area, Kasna, Greater Noida, Uttar Pradesh. The organization was started in 1973 as an unorganized business by Mr. Anil Kumar. However, later it got its registry as a company under MSME. Now, it is jointly managed by Mr. Anil Kumar and his son Vaibhav. The organization manufactures precision metal components used in domestic gas stove assemblies and supplies them to appliance manufacturers and industrial distributors.

Its primary products include brass burner rings, cast iron pan supports and gas valve assemblies. The manufacturing process begins with the procurement of raw materials such as brass ingots, cast iron, aluminium and sheet metal, followed by machining, quality inspection and dispatch of finished components.

As a manufacturing organization operating with substantial investments in machinery and working capital, efficient production planning is essential for maintaining continuous operations, controlling inventory and ensuring timely fulfilment of customer orders. Operational records are currently maintained manually through factory registers, providing an opportunity to transform historical business data into meaningful managerial insights through systematic data analysis.

3. Problem Statement (50-75 Words)

- 3.1. To evaluate whether existing production planning decisions effectively align raw material procurement with actual production requirements.
- 3.2. To investigate the relationship between production output, customer demand and operational performance in order to identify factors contributing to production planning inefficiencies.
- 3.3. To develop a data-driven production planning framework that supports better material utilization, improves operational efficiency and enhances production decision-making.

4. Background of the Problem (200 Words)

Production planning plays a central role in manufacturing organizations because it determines how effectively raw materials, manufacturing resources and customer demand are coordinated throughout the production process. Decisions related to procurement, production scheduling and order fulfilment directly influence inventory levels, working capital utilization and operational efficiency.

Preliminary examination of MetaFab India's operational records indicates that production, procurement and sales information are maintained independently through manual registers. Although these records capture valuable business information, they have not been systematically integrated to evaluate whether procurement decisions adequately support production requirements or whether production output aligns with customer demand.

Without such analysis, management may find it difficult to identify operational inefficiencies, understand material utilization patterns or evaluate the effectiveness of existing production planning practices. Internal operational factors such as production variability, manufacturing defects and resource availability may further influence production performance, while fluctuations in customer demand may affect production scheduling decisions.

This study therefore aims to investigate these operational relationships using primary business data and develop evidence-based recommendations to improve production planning.

5. Problem Solving Approach (400 Words)

5.1 Methods

This study will adopt a structured Business Data Management approach to evaluate the effectiveness of production planning at the selected manufacturing organization. The primary methodology will be **Material Flow Analysis (MFA)** to examine the movement of raw materials from procurement through production to finished goods and customer dispatch. This method is appropriate because it helps identify bottlenecks, material losses and inefficiencies across the manufacturing process.

Descriptive Analytics will be used to summarize historical operational data and understand procurement, production and sales patterns over the study period. **Trend Analysis** will be performed to identify recurring variations in production output, material consumption and customer demand over time.

Comparative Analysis will be used to compare planned activities with actual operational performance in order to identify deviations affecting production efficiency. Where appropriate, **simple forecasting techniques** will be applied to estimate short-term production requirements based on historical demand patterns and support better production planning decisions. The combined use of these methods will provide a comprehensive understanding of operational performance and enable evidence-based recommendations.

5.2 Data Collection

The project will rely entirely on **primary data** obtained directly from the organization's operational records. The core datasets will include procurement records, production records and sales records, while inventory records, machine downtime and labour attendance will be analysed whenever additional operational explanation is required.

The procurement dataset will include variables such as **purchase date, supplier, raw material, quantity purchased and purchase cost** to analyse procurement patterns and material availability. The production dataset will contain **production date, product type, batch details, production quantity and defective output** to evaluate manufacturing performance and material utilization. The sales dataset will include **invoice date, customer, product, quantity sold and sales value** to examine customer demand and order fulfilment. Where necessary, **opening stock, closing stock, material issued, machine downtime and labour attendance** will be analysed to explain operational variations and identify factors affecting production efficiency.

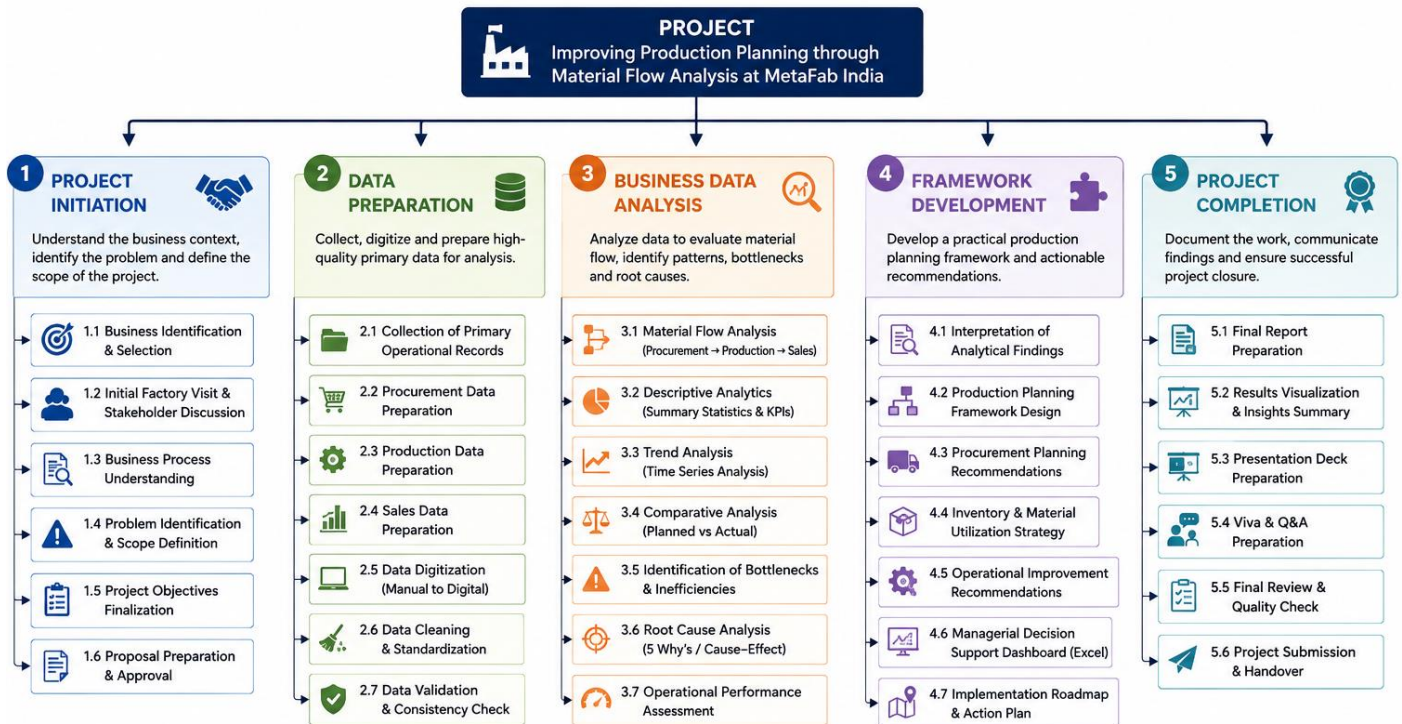
5.3 Analysis Tools

The collected data will be cleaned, organised and analysed using **Microsoft Excel** and **Python**. Microsoft Excel will be used for initial data digitization, validation and basic exploratory analysis because it provides an efficient environment for handling manually maintained operational records. Python, together with the **Pandas** and **NumPy** libraries, will be used for data cleaning, transformation, statistical analysis and material flow evaluation. These tools are selected because they provide reliable, reproducible and scalable methods for analysing operational data and generating evidence-based insights to support managerial decision-making.

6. Expected Timeline

WORK BREAKDOWN STRUCTURE (WBS)

Improving Production Planning through Material Flow Analysis: A Case Study of MetaFab India



PROJECT TIMELINE – GANTT CHART

A Data-Driven Framework for Improving Production Planning through Material Flow Analysis: A Case Study of MetaFab India

ID	ACTIVITY	DURATION (Weeks)	WEEK / DATE						
			Week 1 1 – 7 Jun 2026	Week 2 8 – 14 Jun 2026	Week 3 15 – 21 Jun 2026	Week 4 22 – 28 Jun 2026	Week 5 29 Jun – 5 Jul 2026	Week 6 6 – 12 Jul 2026	Week 7 13 – 19 Jul 2026
1	Business Selection & Data Collection	Completed							
2	Proposal Preparation	1 Week							
3	Data Digitization & Validation	1 Week							
4	Exploratory Data Analysis	1 Week							
5	Material Flow & Operational Analysis	1 Week							
6	Framework Development	1 Week							
7	Final Report Preparation	1 Week							
8	Presentation & Viva Preparation	1 Week							

Legend: Completed In Progress / Upcoming Not Started

7. Expected Outcome (Briefly explain in 150-200 words)

- 7.1. The project is expected to provide MetaFab India with a structured understanding of its existing production planning practices and the operational relationships between procurement, production and customer demand.
- 7.2. The proposed production planning framework will enable management to make more informed procurement decisions, improve material utilization, monitor production performance and identify operational bottlenecks affecting manufacturing efficiency.
- 7.3. By integrating business data from multiple operational functions, the project aims to transform manually maintained production records into actionable managerial insights. The recommendations developed through this study are expected to support more efficient resource utilization, improve production planning decisions and strengthen overall operational performance while remaining practical and implementable within the existing business environment.